

Rakibul Hasan

PhD Researcher | AI & IoT for Smart Energy Efficiency Systems

+33 7 58 00 42 72 rakibul.hasan@u-pec.fr rakibs8583@gmail.com France

LinkedIn | Research Gate | Google Scholar | ORCID

OBJECTIVE

PhD Researcher at the CERTES Laboratory, Université Paris-Est Créteil (UPEC), specializing in the development of AI- and IoT-based intelligent systems for energy efficiency and smart building management. My research focuses on data-driven solutions that optimize energy consumption, detect anomalies, and support regulatory compliance. Core expertise includes artificial intelligence, IoT sensor networks, and sustainable energy forecasting and regulatory compliance.

EDUCATION

Université Paris-Est Créteil (UPEC) Oct 2025 – Present

PhD Researcher - AI and IoT for Smart Energy Efficiency Systems Créteil, France

- **Research Focus:** Proactive energy consumption and efficiency management in buildings using Artificial Intelligence and the Internet of Things (IoT).
- **Laboratory:** Centre d'Études et de Recherche en Thermique, Environnement et Systèmes (CERTES).
- **Supervisors:** Prof. Gilles Lefebvre, Dr. Mahamadou Abdou Tankari, Dr. Jura ARKHANGELSKI

Free University of Bozen-Bolzano Apr 2026 – Jul 2026

Visiting Doctoral Researcher (PhD International Mobility) Bolzano, Italy

- **Objective:** Develop an "Edge-Cloud AI" architecture for smart building energy management to address limitations of centralized cloud-based IoT platforms, such as high latency and bandwidth inefficiency.
- **Collaboration:** Working with the Intelligent Networks and Systems group at the Free University of Bozen-Bolzano, under the supervision of Prof. Antonio Liotta and Dr. Muhammad Azfar Yaqub.

Université Paris-Est Créteil (UPEC) Sep 2023 – Aug 2025

M1, M2. – International Master in Optics, Image, Vision, Multimedia Créteil, France

- **Specialization:** International Biometrics and Intelligent Vision.
- **Key Courses:** AI Innovation, Computer Vision, Machine Learning, Pattern Recognition, Biometrics, Software Integration, Ethics & Privacy, Project Management.

Shanto-Mariam University of Creative Technology Jun 2018 – Jun 2022

B.Sc. in Computer Science and Engineering Dhaka, Bangladesh

- **Specialization:** Artificial Intelligence and Data Processing.
- **Key Courses:** Machine Learning, Digital Signal Processing, Artificial Intelligence, Data Structures, Database Systems, Software Engineering.

RESEARCH INTERESTS

- Smart Building Efficiency
- Time-Series Forecasting
- AI for Sustainability
- Hybrid Deep Learning
- Anomaly Detection
- Regulatory Compliance
- IoT Sensor Networks
- Peak Shaving Strategies

RESEARCH & PROFESSIONAL EXPERIENCE

CERTES Laboratory, Université ParisEst Créteil (UPEC) Oct 2025 – Present

Doctoral Researcher (Contrat Doctoral) Créteil, France

- Designing proactive energy management systems using IoT sensors and AI.
- Developing hybrid deep learning models (LSTM + Interpolation) to handle irregular time-series data.
- Analyzing building load profiles to implement "Peak Shaving" strategies.
- Ensuring compliance with French environmental regulations (Décret Tertiaire) through data auditing

CERTES Laboratory, UPEC Feb 2025 – Sep 2025

R&D Research Intern (Master 2 Thesis) Créteil, France

- Developed an innovative approach using Akima Spline Interpolation combined with LSTM networks.

- Processed real-time IoT sensor data to enhance anomaly detection and system monitoring.
- Implemented Artificial Intelligence models for failure prediction and reliability assessment.
- Achieved significant improvement in energy forecasting accuracy for smart buildings compared to traditional methods.
- Published findings in an IEEE International Conference

Micco, STATION F

Machine Learning Intern

Jun 2024 – Sep 2024

Paris, France

- Fine-tuned Large Language Models (LLMs) for NLP applications to improve accuracy.
- Developed predictive models for context-aware automation.
- Optimized data processing pipelines for scalable AI-driven applications.

Smart Technologies Ltd

Associate IT Security

Jun 2022 – Jul 2023

Dhaka, Bangladesh

- Developed AI-powered security tools for threat detection and prevention.
- Monitored network traffic patterns to identify anomalies using ML algorithms.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, MATLAB, C++, HTML, CSS, PyTorch, TensorFlow
- **Machine Learning & Deep Learning:** Transformer, LSTM, GRU, CNNs, Autoencoders, XGBoost, Contrastive Learning, Sparse Representations, Model Explainability, Representation Learning
- **Data Science:** Python (Pandas, NumPy, Scikit-learn), Time-Series Forecasting, Data Visualization.
- **IoT & Energy:** Smart Sensors, BEMS (Building Energy Management Systems), Energy Optimization.
- **Research & Collaboration:** Git, Docker, LaTeX, SQL, VS Code.

PUBLICATIONS

- **R. Hasan**, M. Mansouri, J. Arkhangelski, and M. Abdou Tankari, "Interpretable Multi-Sensor Fusion for Short-Term Energy Consumption Forecasting," *Energies*, vol. 19, no. 9, p. 2230, May 2026, doi: 10.3390/en19092230
- J. Arkhangelski, **R. Hasan**, M. Abdou Tankari, and G. Lefebvre, "Benchmarking Deep Learning Architectures for 24-Hour Energy Forecasting in Smart Buildings Using Real-World IoT Data," *Artificial Intelligence Research and Applications*, vol. 1, no. 2, 2026.
- J. Arkhangelski, **R. Hasan**, M. Abdou Tankari, and G. Lefebvre, "Comparative Evaluation of Deep Learning-Based AI Models for 24-Hour Energy Forecasting in Smart Buildings Using IoT Sensor Data," in *Proc. 2025 13th International Conference on Smart Grid (icSmartGrid)*, Glasgow, United Kingdom, May 2025, doi: 10.1109/icSmartGrid66138.2025.11071776

HOBBIES & INTERESTS

Tech Conferences and Meetups, Open Source Contributions, Photography, Exploring New Technologies, Problem-solving, Traveling, Serving with a Smile, Reading Books, Blogging, Volunteering, Sports.

LANGUAGES

- **English:** Full Professional Proficiency
- **French:** Intermediate

REFERENCES

- **Dr. Mahamadou Abdou-Tankari** | CERTES Laboratory, UPEC | mahamadou.abdou-tankari@u-pec.fr | PhD Thesis Director
- **Prof. Antonio Liotta** | UNIBZ | antonio.liotta@unibz.it | Supervisor (PhD International Mobility)
- **Dr. Jura Arkhangelski** | CERTES Laboratory, UPEC | jura.arkhangelski@u-pec.fr | PhD Thesis Supervisor
- **Prof. Gilles Lefebvre** | CERTES Laboratory, UPEC | gilles.lefebvre@u-pec.fr | PhD Thesis Co-Director
- **Prof. Amine Nait-Ali** | LISSI Laboratory, UPEC | amine.naitali@u-pec.fr | Master's Program Director